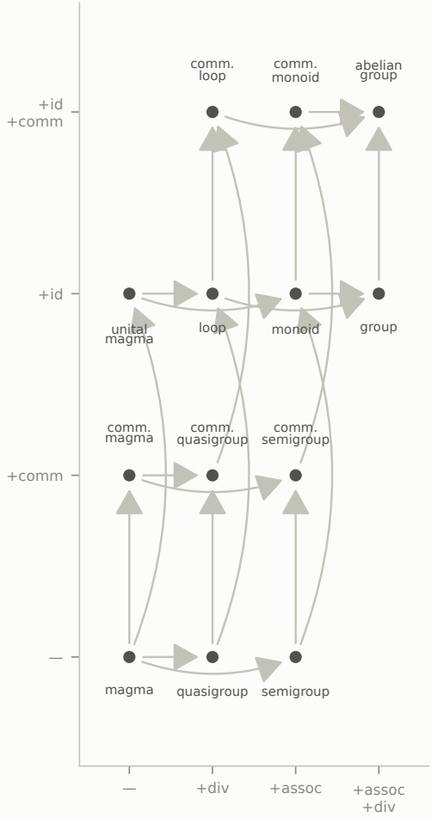
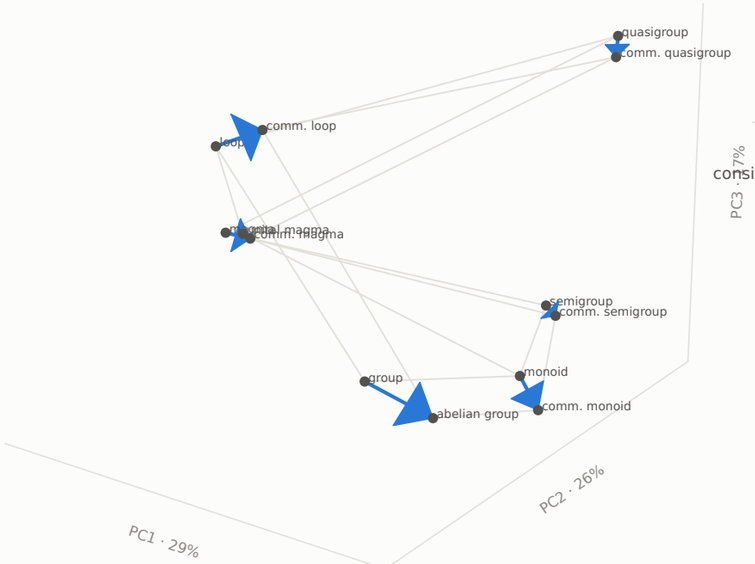
1 · Structures are sets of axioms

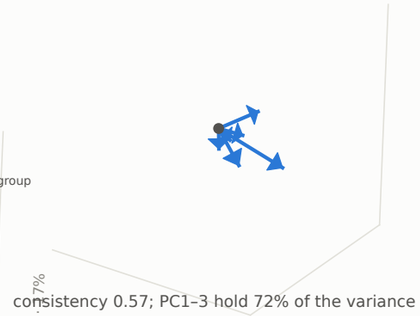


2 · Adding an axiom is an arrow + commutativity in Qwen3.5-0.8B (top-3 PCs)



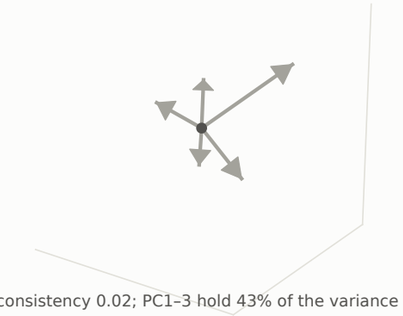
3 · Line the arrows up

model



consistency 0.57; PC1–3 hold 72% of the variance

random-init twin

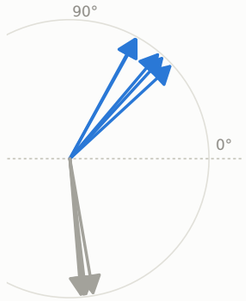


consistency 0.02; PC1–3 hold 43% of the variance

4 · Measure in the full space: angle of each arrow to the others' mean

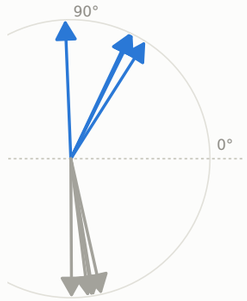
model above the dashed line (blue), twin below (grey); 0° = same direction, 90° = unrelated

+ associativity



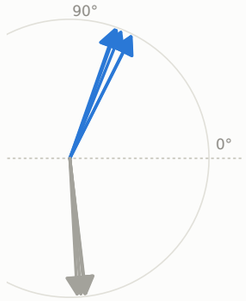
consistency 0.47 (p = 0.003)
twin 0.06

+ identity



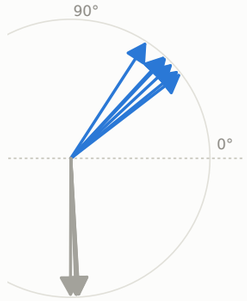
consistency 0.22 (p = 0.057)
twin 0.07

+ divisibility



consistency 0.26 (p = 0.043)
twin 0.05

+ commutativity



consistency 0.57 (p < 0.001)
twin 0.02